

Apache Iceberg and Trino: Powering Data Lakehouse Architecture

Apache Iceberg is a cornerstone of any open data lakehouse, providing the transactional foundation upon which highly scalable and flexible analytics can flourish. Along with Trino, it can be used to build a robust, scalable, and high-performance data lakehouse.



Over the past ten years, the emergence of Big Data has transformed how organisations store and process their data. The performance and reliability of traditional data warehouses lacks flexibility and cost-effectiveness. At the same time, data lakes have scale and affordability but are challenged with governance, schema enforcement, and performance limitations.

Enter the data lakehouse -- a new data architecture that combines the scale-out store of data lakes with the transactional and governance features of data warehouses. By allowing SQL workloads natively on object storage with

Data lake vs data warehouse vs data lakehouse

Feature	Data warehouse	Data lake	Data lakehouse
Storage format	Proprietary	Open (Parquet, ORC)	Open (Parquet, ORC)
Cost	High	Low	Moderate
Data types supported	Structured	Semi/Unstructured	All
Performance	High	Low	High
Governance	Strong	Weak	Strong
Schema enforcement	Rigid	Loose	Flexible
ACID compliance	Yes	No	Yes